Function as a Binary Relation

Cartesian Product. The Cartesian product of two sets A and B is defined as

$$A \times B = \{(a, b) \mid a \in A, b \in B\}$$

- (a, b) is called the ordered pair; a is its first component and b the second component.

Example 1. The Cartesian product of $A = \{a, b, c\}$ and $B = \{1, 2, 3, 4\}$ is

$$\begin{aligned} A \times B &= \{(a,1),(a,2),(a,3),(a,4),(b,1),(b,2),(b,3),(b,4), \\ &\quad (c,1),(c,2),(c,3),(c,4)\}. \end{aligned}$$

□

Binary Relation. Any subset R of $A \times B$ is called a binary relation from A to B.

- The set of first components of the elements of R is called the domain of R, $dom(R)$.
- The set of second components of the elements of R is called the range of R, $range(R)$.

Example 2. Consider $A = \{a, b, c\}$, $B = \{1, 2, 3, 4\}$, and $A \times B = \{(a, 1), (a, 2), (a, 3), (a, 4), (b, 1), (b, 2), (b, 3), (b, 4), (c, 1), (c, 2), (c, 3), (c, 4)\}$. The following are binary relations from A to B.

$$\begin{aligned} R_1 &= \{(a,1),(b,2),(c,2)\} \\ R_2 &= \{(a,3),(b,3),(c,3),(c,4)\} \\ R_3 &= \{(a,4),(b,4)\} \\ R_4 &= \{(a,3),(b,1),(b,2),(c,1),(c,3)\} \end{aligned}$$

□

Function. A binary relation R is a function from A to B if

1. $dom(R) = A$
2. No two elements of R have the same first components.

Example 3. The following are functions from $A = \{a, b, c\}$ to $B = \{1, 2, 3, 4\}$.

$$\begin{aligned} R_1 &= \{(a,1),(b,2),(c,2)\} \\ R_2 &= \{(a,3),(b,3),(c,3)\} \\ R_3 &= \{(a,4),(b,3),(c,1)\} \end{aligned}$$

□

Example 4. The following are not functions from $A = \{a, b, c\}$ to $B = \{1, 2, 3, 4\}$.

- $R_1 = \{(a, 1), (b, 2)\}$ ($dom(R_1) = \{a, b\} \neq A$)

- $R_2 = \{(a, 1), (b, 2), (c, 2), (c, 4)\}$ (Two elements, $(c, 2)$ and $(c, 4)$, have the same first components.)

- $R_3 = \{(b,2),(c,2),(c,3)\}$ (Neither the domain of R_3 is A nor its elements have distinct first components.)

$\square$

Remark 1. If a binary relation is a function from A to B, then it is usually denoted by f, and is generally represented as $f : A \to B$ (read "f is a function from A to B").

Practice Problems.

1. Construct a function f from $A = \{1,2,3\}$ to $B = \{1,2,\ldots,100\}$ such that the second component of each element of f is the square of its first component. Also, find the range of f.
2. Construct a function f from $A = \{1,2,3\}$ to $B = \{1,2,\ldots,100\}$ such that the second component of each element of f is two more than the cube of its first component. Also, find the range of f.

(**I shall welcome your suggestions to improve these notes.**)

Function. A *function* from a set X to a set Y is a rule f which assigns to ever element x of X a unique element y of Y.

- The set X is called the *domain* of f and the set Y is called the *codomain* of f.
- y is called *image* of x under f, and is usually represented in terms of a formula $y = f(x)$.
- The set of all images is called the *range* of f.
- The variable x which represents all the elements of the domain of f is called the *independent variable.*
- The variable y which represents all the elements of the range of f is called the *dependent variable.*
- If $range(f) = Y$, the full codomain, then f is said to be an *onto* function.
- f is said to be a *one-to-one* function if distinct elements of domain has distinct images in codomain.

Note.

1. Domains and ranges of all our functions will be subsets of real numbers.
2. The domain of f is all real numbers if there is no square root or fraction in the formula.

Geometric Approach.

- **Vertical Line Test:** Every vertical line intersects the graph of a function f exactly at one point.
- If a horizontal line intersecting the graph meets the y-axis at the point y, then y belongs to the range of f. The set of all such y points form the range of f.
- f is onto if every horizontal line intersects the graph of f.
- f is one-to-one if every horizontal line intersects the graph of f exactly at one point.

Example 1. Sketch the function $f(x) = x^2$ and find its domain and range. Check wether it is one/onto or not.